1.8 Reaction Kinetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Easy

Time allowed: 30

Score: /22

Percentage: /100

Question 1a

This question is about effective reactions and reaction rates.

According to collision theory, state **two** conditions that particles need for an effective collision to occur.

[2 marks]

Question 1b

An increase in concentration means that there are more particles per unit volume.

In terms of collisions, explain how this increases the rate of reaction.

[1 mark]

Question 1c

Aqueous hydrogen peroxide, H_2O_2, decomposes to form oxygen and water.

$$2H_2O_2\ (aq) \rightarrow 2H_2O\ (l) + O_2\ (g)$$

A student investigates the decomposition of a hydrogen peroxide solution as outlined below.

- The student adds 25.00 cm^3 of H_2O_2 (aq) to a conical flask.
- The student adds a small spatula measure of manganese dioxide, MnO_2.
- They quickly connect the flask to a gas syringe.
- The student measures the volume of oxygen every 100 seconds.

i)

State the role of the manganese dioxide in this reaction.

[1]

The student obtained the following results.

Time / s	0	100	200	300	400	500	600	700
Volume of O_2 / cm^3	0.0	7.0	13.5	18.0	21.0	22.5	23.0	23.0

ii)

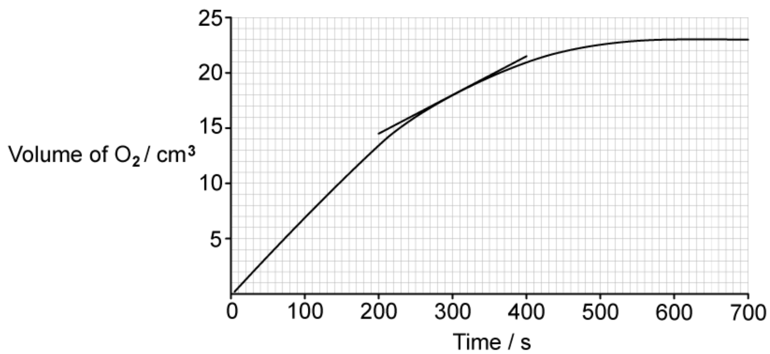
Calculate the average rate of reaction, in $cm^3\ s^{-1}$, for the first 100 seconds of the reaction.

[1]

[2 marks]

Question 1d

The student plotted a graph of volume of O_2 against time.



Use the student's graph and tangent to find the rate of the reaction, in $cm^3\ s^{-1}$, at t = 300 s.

Show your working.

[3 marks]

Question 2a

This question is about the following general reaction

$$A + B \rightleftharpoons C + D$$

The reaction pathway diagram for the reaction is shown in Fig. 2.1.

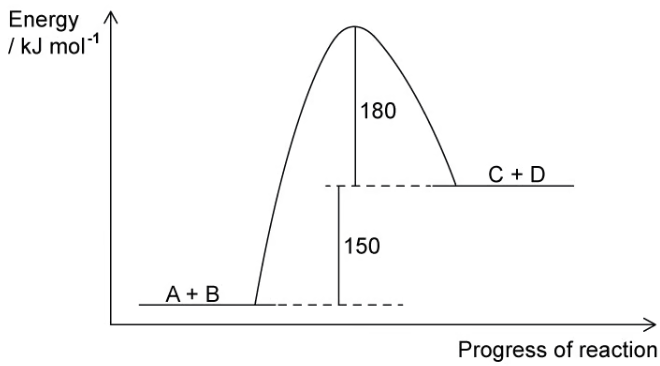


Fig. 2.1

Explain whether the forward reaction is exothermic or endothermic.

[2 marks]

Question 2b

Define the term activation energy.

[1 mark]

Question 2c

Use Fig. 2.1 to calculate the activation energy for the forward and backward reactions.

[2 marks]

Question 2d

Explain, using Fig. 2.2, how the addition of a catalyst affects the rate of reaction.

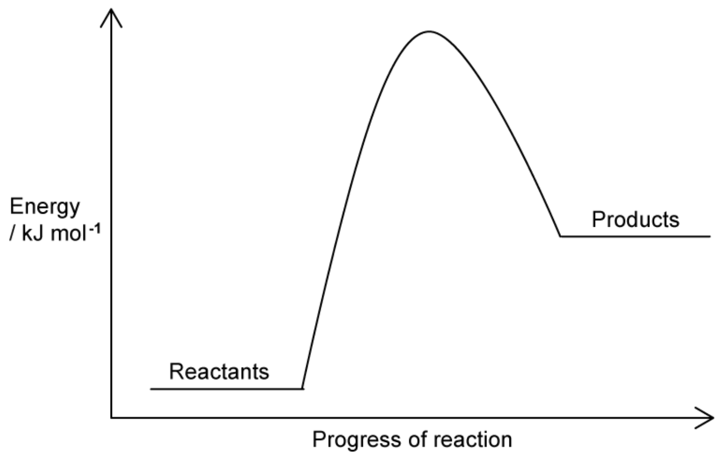


Fig. 2.2

[3 marks]

Question 3a

This question is about Boltzmann distribution curves.

The Boltzmann distribution curve of molecular energies for a general reaction at a given temperature is shown in Fig. 3.1.

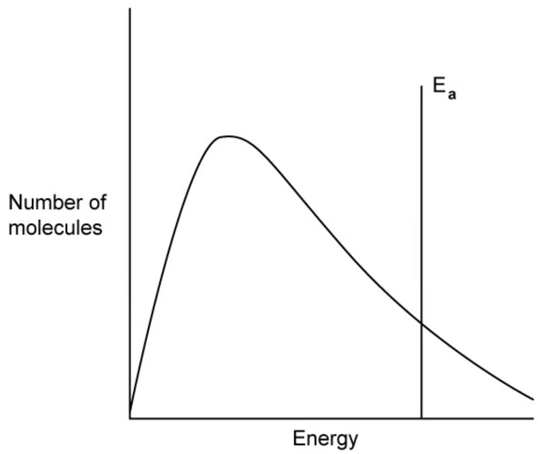


Fig. 3.1

State what will happen to the curve when the temperature of the reaction is decreased.

[2 marks]

Question 3b

Comment on how, if at all, the total area under a Boltzmann distribution curve changes with temperature.

[1 mark]

Question 3c

State what happens to the Boltzmann distribution curve in Fig. 3.1 when a catalyst is added to the reaction.

[2 marks]

Question 3d

The Boltzmann distribution of energies for a gas is shown in Fig. 3.2.

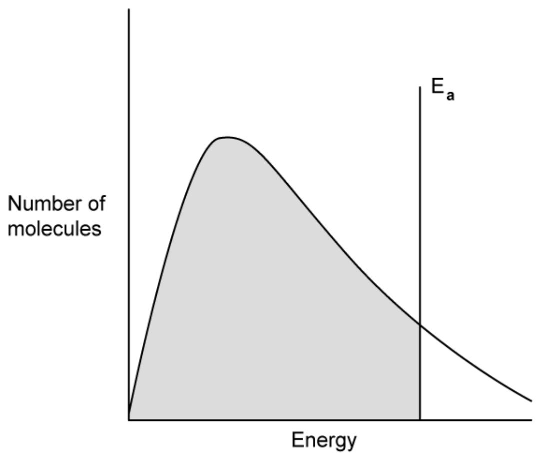


Fig. 3.2

State what the shaded area of Fig. 3.2 represents.

[1 mark]

Model Answers: Easy

1a

a) According to collision theory, the **two** conditions that particles need for an effective collision are:

- Sufficient energy
  OR
  Energy above the activation energy / $E > E_a$; [1 mark]
- (To be in the) correct orientation; [1 mark]

[Total: 2 marks]

- For a chemical reaction to occur, the particles must collide in the correct orientation and with enough energy

1b

b) In terms of collisions, increasing the concentration increases the rate of reaction by:

- Increasing the number of (effective / successful) collisions per unit time
  OR
  Increase the frequency of (effective / successful) collisions
  OR
  Having particles collide (effectively / successfully) more often; [1 mark]

[Total: 1 mark]

- The idea of the frequency of (effective / successful) collisions is one of the most commonly missed marks in an exam
- **TIP:** Adding the phrase ***more frequent effective collisions*** should be sufficient to score 2 common exam marks when the rate of reaction increases
  - Obviously, you should write ***less*** *frequent effective collisions* if that is appropriate to the question you are answering

1c

c)

i) The role of the manganese dioxide in this reaction is:

- Catalyst
  OR
  To increase the rate of reaction / to speed the reaction up; [1 mark]

ii) The average rate of reaction, in $cm^3\ s^{-1}$, for the first 100 seconds of the reaction is:

- $\left(\frac{7}{100}\right)$ = 0.07 ($cm^3\ s^{-1}$); [1 mark]

[Total: 2 marks]

- The manganese dioxide is not involved in the chemical equation so it is not a reactant or an oxidising / reducing agent
- Manganese dioxide is not an acid as it does not have hydrogen ions
- Manganese dioxide is not an alkali as it does not dissolve in water to release hydroxide ions
- The only realistic role left for manganese dioxide is as a catalyst
- It is rare that you would be asked to use results directly from a table to calculate the rate, you would usually be asked to plot the graph and then use that to calculate the rate
- **Remember:** If you aren't sure what to calculate, the units help you $cm^3\ s^{-1}$ or cm^3 per second
  - This is telling you to have a value for the volume and divide it (per) by the time

1d

d) To find the rate of the reaction, in $cm^3\ s^{-1}$, at t = 300 s

- Volume at 200 s = 14.5 (cm^3)
  AND
  Volume at 400 s = 21.5 (cm^3); [1 mark]
- Difference in volume = 21.5 - 14.5 = 7 (cm^3); [1 mark]
- Rate $\left(= \frac{7}{200}\right)$ = 0.035 ($cm^3\ s^{-1}$); [1 mark]

[Total: 3 marks]

- If your final answer is 0.035 cm^3 s^{-1} then all three marks would be awarded, even without working
    - For the second mark, other values can be read off the tangent as long as they lead to the same final answer
    - Rates in the range of 0.0325 - 0.0375 cm^3 s^{-1} would be accepted for the final mark
- In an exam, you could be expected to plot the graph, add the tangent and then calculate the rate
- **Tip:** Extend your tangent to points that will be easy to read off
- Examiners will usually have a range of acceptable answers because the position of the tangent will be slightly different across all students

2a

a) The forward reaction is:

- Endothermic; [1 mark]
- (Because,) the energy level of the products is higher / greater than / above the energy level of the reactants; [1 mark]

[Total: 2 marks]

- Endothermic reactions take energy in, which results in the products having a higher energy level than the reactants
    - An alternative acceptable answer is that the products have taken in / gained energy but the responses in the above mark scheme are stronger answers
- Make sure you know your reaction pathway diagrams **correctly**
    - Examiners do use these to introduce longer questions and comment on how the problems that anything exothermic and endothermic causes for students

2b

b) Activation energy is:

- The minimum amount of energy required for a collision to be effective / for a reaction to take place; [1 mark]

[Total: 1 mark]

- This is another definition for you to know
- Although, this is not a definition that is commonly asked for because questions tend to focus on:
  - Calculating values for activation energy
  - Using collision theory to explain the effects of changing temperature in relation to activation energy

2c

c) The activation energy values for the forward and backward reactions are:

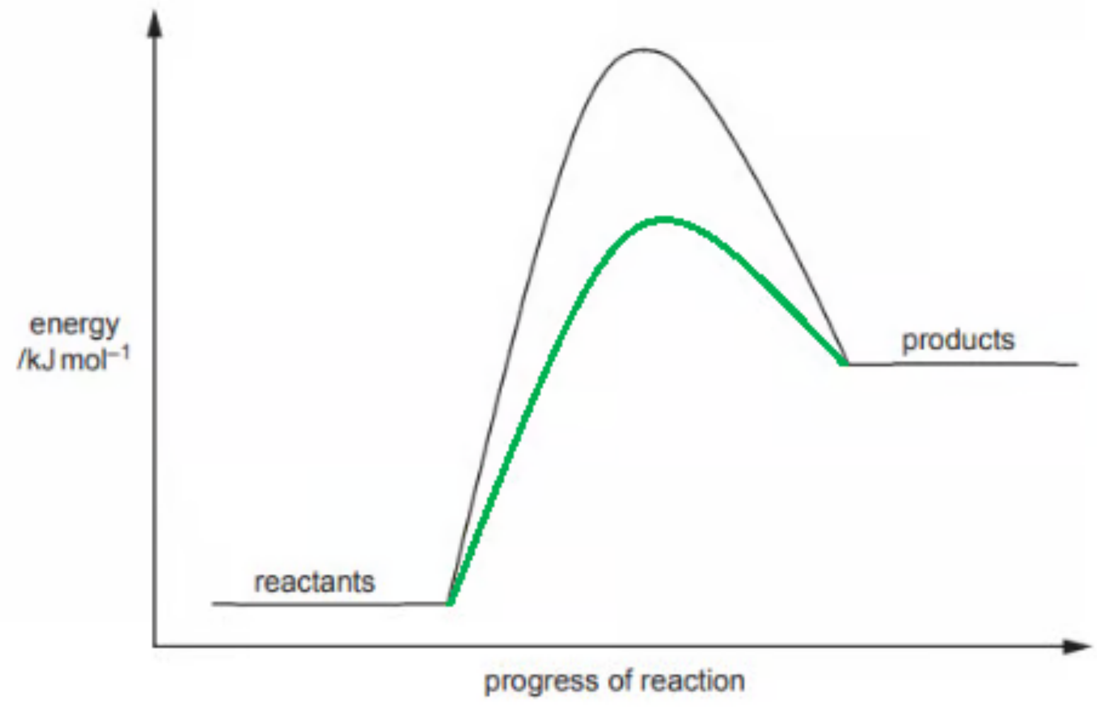
- Forward reaction = 150 + 180 = 330 (kJ mol^{-1}); [1 mark]
- Backward reaction = 180 (kJ mol^{-1}); [1 mark]

[Total: 2 marks]

- The activation energy for the forward reaction is the energy required from the reactants to the peak of the curve = 150 + 180 = 330 kJ mol^{-1}
- The activation energy for the reverse reaction is the energy from the products to the peak of the curve = 180 kJ mol^{-1}
- In reversible reactions, the endothermic reaction will always have a higher activation energy than the exothermic reaction
- The activation energy value for the endothermic value should equal the enthalpy change + the activation energy of the exothermic reaction

2d

d) The addition of a catalyst affects the rate of reaction by:



- Catalyst curve below the original curve
  AND
  The peak must be above the energy level of the products; [1 mark]
- Providing an alternative pathway / route for the reaction; [1 mark]
- That requires less / lower activation energy; [1 mark]

[Total: 3 marks]

- How a catalyst works is a common question, including adding the catalyst curve to an already existing enthalpy profile diagram
- Common mistakes include:
  - Not mentioning alternative pathway **AND** lower activation energy
  - Unnecessarily talking about how a catalyst is not used / used up in a reaction
  - Unnecessarily talking about how heterogeneous catalysts work, e.g. the adsorb, react, desorb process
  - Forgetting to make sure that the catalyst curve has a "hump" for endothermic reactions

3a

a) The effect of decreasing the temperature on the curve is:

- (The curve will) shift to the left

AND

Be higher / move up; [1 mark]

- Curve finishes below the level of the original curve; [1 mark]

[Total: 2 marks]

- Not mentioning alternative pathway **AND** lower activation energy
- Unnecessarily talking about how a catalyst is not used / used up in a reaction
- Unnecessarily talking about how heterogeneous catalysts work, e.g. the adsorb, react, desorb process
- Forgetting to make sure that the catalyst curve has a "hump" for endothermic reactions

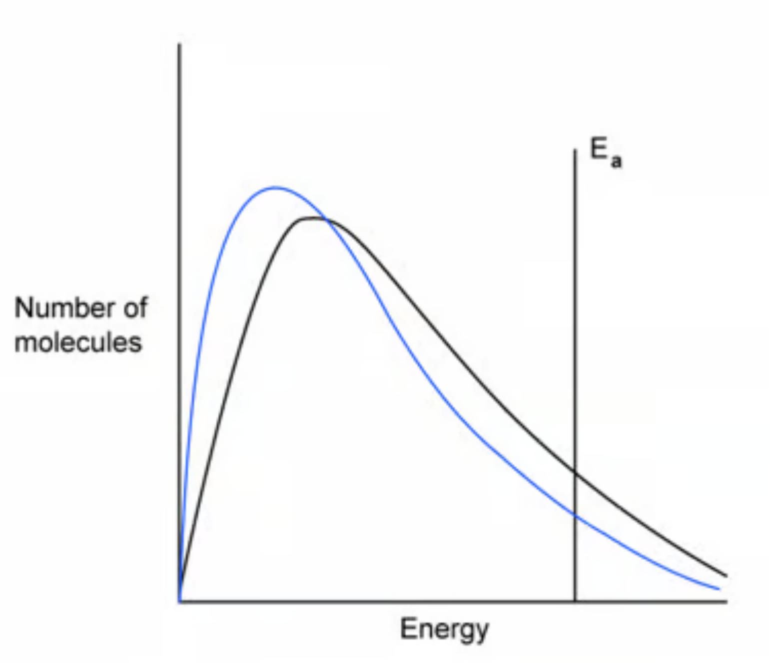
3a

a) The effect of decreasing the temperature on the curve is:

- (The curve will) shift to the left
  AND
  Be higher / move up; [1 mark]
- Curve finishes below the level of the original curve; [1 mark]

[Total: 2 marks]

- Lower temperatures mean that the proportion of molecules with an energy greater than the activation energy, E_a, is less
  - This results in the curve being higher and to the left
- **Careful:** It also results in the curve finishing below the original curve



- Make sure you read the question carefully because many students make mistakes by answering about the temperature increasing when the question asks about decreasing

3b

b) With changing temperature, the total area under a Boltzmann distribution curve:

- Does not change / remains the same; [1 mark]

[Total: 1 mark]

- The total area under the curve represents the number of molecules within the sample
- The total area under the same remains the same, as the sample has the same number of molecules in, regardless of temperature

3c

c) When a catalyst is added to the reaction:

- The curve will stay the same; [1 mark]
- The line for E_a / activation energy will shift to the left; [1 mark]

[Total: 2 marks]

- It is important that you know how Boltzmann distribution curves can be affected
- The addition of a catalyst has no effect at all on the shape of the curve, it just shifts the activation energy line to the left
  - This matches the idea of a catalyst providing an alternative route for the reaction of lower activation energy

3d

d) The shaded area of Fig. 3.2 represents:

- The number of molecules that do not have enough energy to react
  OR
  The number of molecules with energy less than or equal to E_a; [1 mark]

[Total: 1 mark]

- Boltzmann distribution curve questions often focus on the non-shaded area because this represents the number of molecules / particles with enough energy to react

- Mark schemes will often show the energy less than or equal to E_a as $E \leq E_a$
- **Tip:** If you are not sure about your < and > symbols then describe them in words instead